



The Pathways Project

Intelligent document search for pediatric clinical decision support



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Project Goal

Build a pathway-aware RAG chatbot that helps clinicians and staff search Connecticut Children's pediatric clinical pathways with fast, relevant, and source-grounded responses.

Problem and Use Case

Clinical pathways are dense, multi-document resources. The project focuses on turning guideline libraries into a searchable assistant that keeps answers tied to the selected pathway.

67

PATHWAYS

30

PAGES / PATHWAY

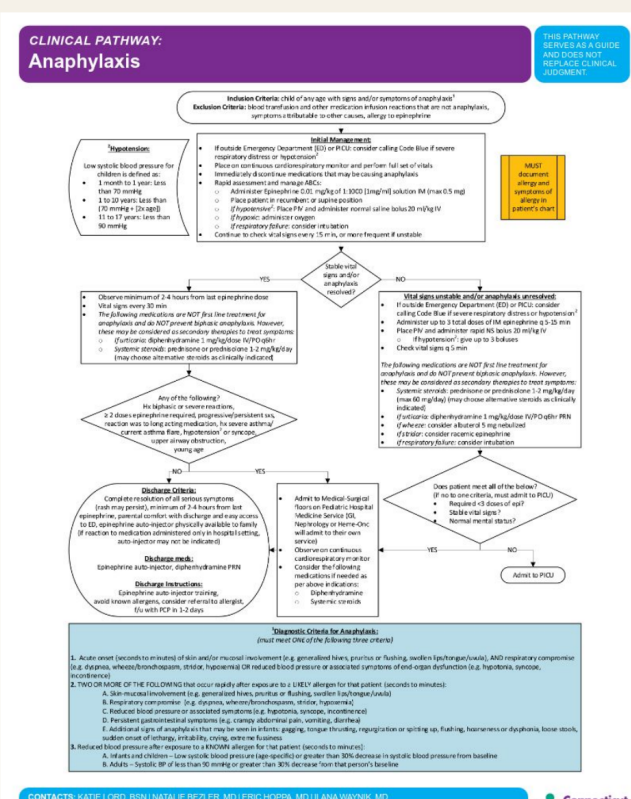
2010

SOURCE PAGES

Why this matters

- Clinical users need quick navigation across algorithms, educational modules, and related pathway PDFs.
- General-purpose LLM answers are not enough; retrieval must be limited to the selected clinical context.
- The system should reduce document-hunting while preserving clinical judgment and source visibility.

Clinical Pathway Source



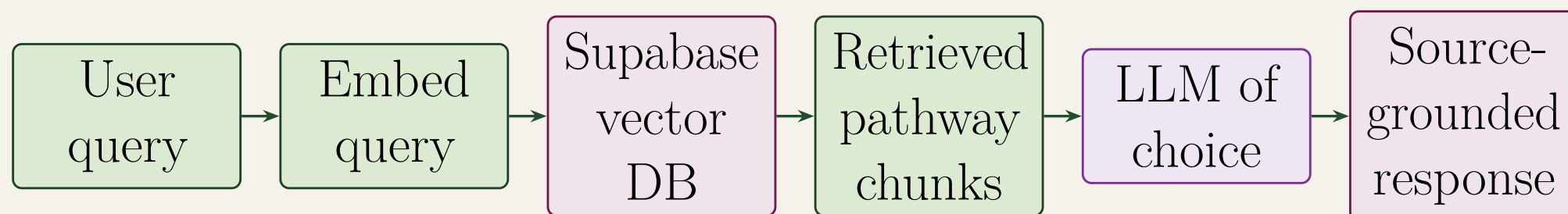
Anaphylaxis pathway: triage, treatment, and disposition algorithm.

Success Criteria

- Search is scoped to a chosen pathway, not the open web.
- Retrieval returns precise chunks with pathway and page metadata.
- Answers are presented alongside their source context.
- Monitoring and feedback enable iterative quality improvement.

Baseline RAG Pattern

Retrieval augmented generation turns a user query into a vector search over embedded documents, then gives the LLM the retrieved context before it answers.



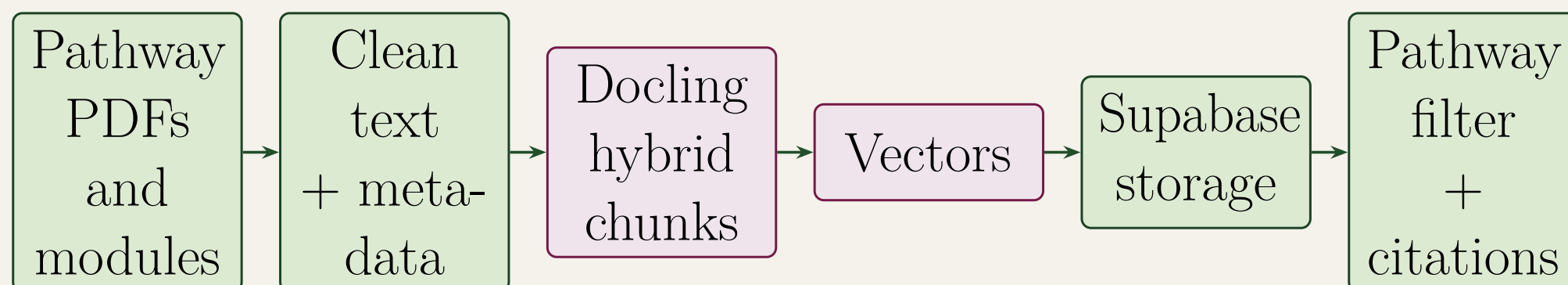
Query embedding → vector search → retrieved context → response generation.

Why not just a general-purpose LLM?

- A general LLM hallucinates specifics; clinical answers must be traceable to a pathway document.
- Open-web retrieval returns irrelevant or outdated guidance instead of the hospital's own pathway.
- Ground-truth sources at CT Children's live in versioned PDFs, not in the pre-training corpus.

Heavy-Digestion Pipeline

Raw PDFs carry broken tables, split paragraphs, and embedded figures that a direct ingest would dump into the index as noise. The heavy-digestion pipeline does the expensive document work *once*, up front, so retrieval at query-time is fast and precise.



Document transformation → chunking → embedding → vector storage → pathway filtering.

Design Principle

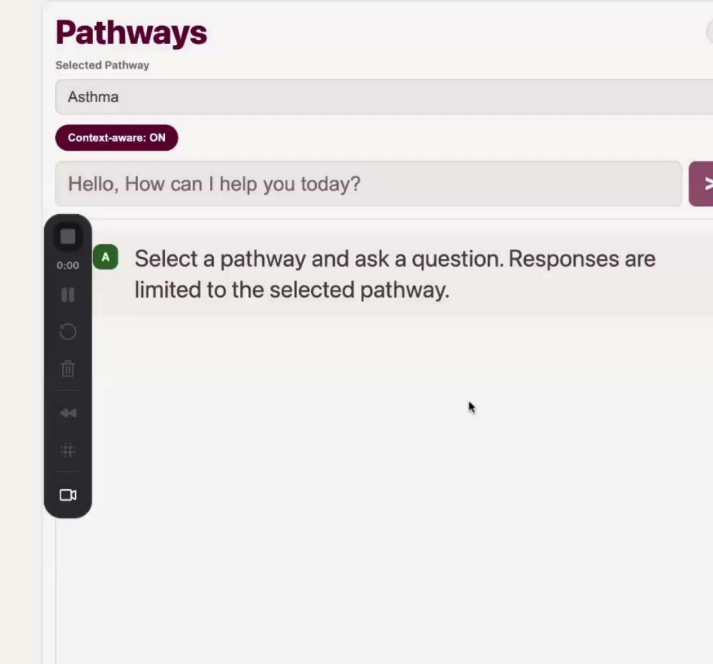
Do more document work *before* retrieval, so the chatbot can do less guessing *during* clinical question answering.

Direct Ingest vs. Heavy Digest

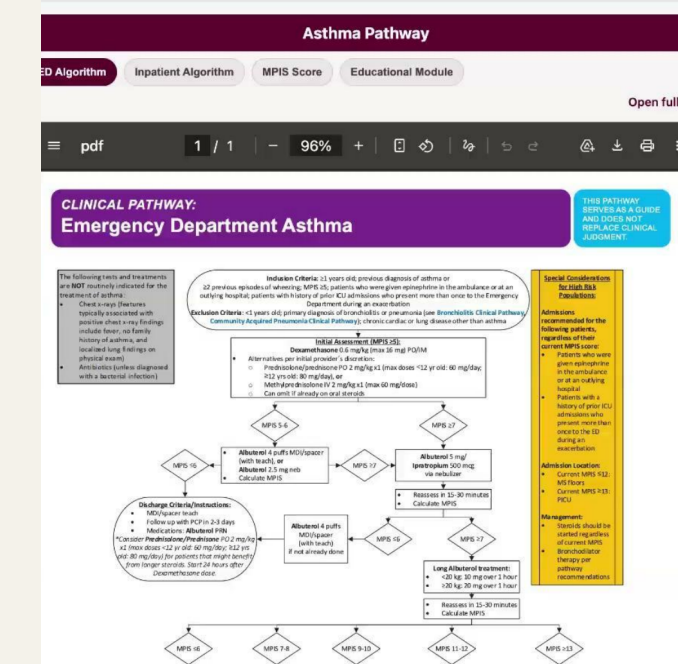
dimension	Direct ingest	Heavy digest (ours)
Chunking	fixed char windows	document-aware (~512 tok)
Metadata	none / filename	pathway, section, page
Retrieval	whole corpus	pathway-filtered
Fidelity	lossy, generic	traceable, specific

Prototype Interface

The demo lets a user select a pathway, ask a question, and inspect the source PDF in one workflow.



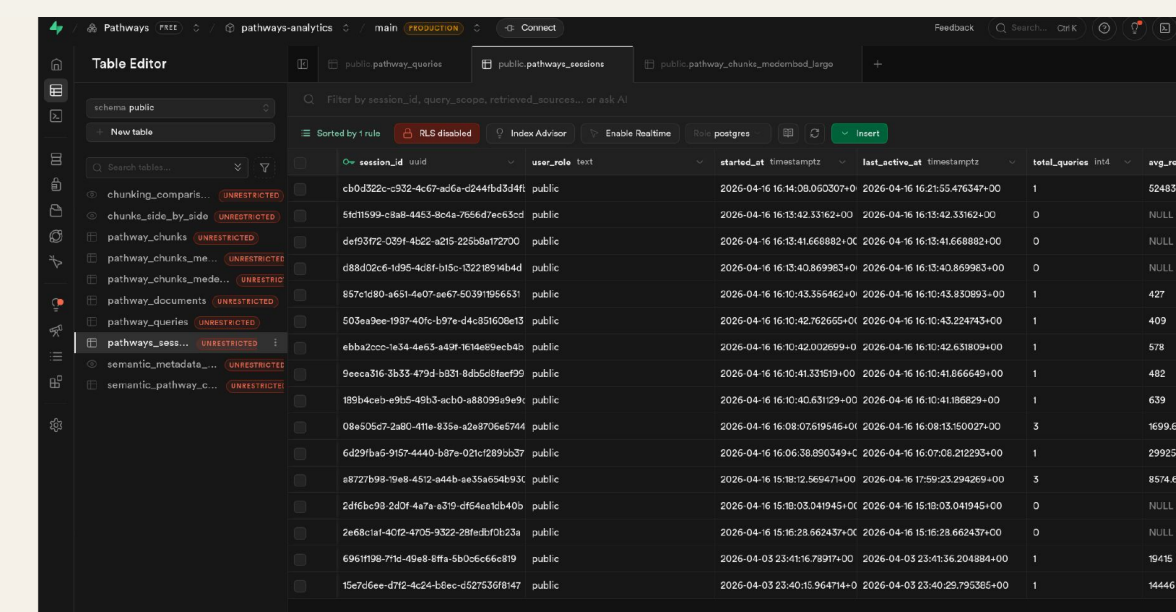
Pathway-scoped chat, context-aware mode



Embedded PDF viewer, tabbed modules

- Pathway selection constrains retrieval to relevant source material.
- Context-aware mode supports natural follow-up questions.
- PDF tabs expose ED/inpatient algorithms, scores, and educational modules.

Monitoring and Future Improvements



Supabase session table: per-query timestamps, user ids, pathway ids, latencies.

- Usage dashboard for session stats, response times, and retrieval behavior.
- Response feedback so clinicians can grade answer quality.
- Lighter preprocessing; continued prompt iteration.

Takeaways

Clinical alignment

communication with CT Children's

Prior art

learning from Seattle and Arizona

Impact

building something that matters

Next step: validate retrieval quality against pathway-specific questions, then connect monitoring and feedback data back into the document-processing pipeline.